

Surgery for a brain tumour

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About surgery for a brain tumour

Surgery is often the main [treatment for a brain tumour](#). The operation you have depends on the size of the tumour and its position. Only highly specialised surgeons called neurosurgeons do brain surgery. It is done in specialist centres or hospitals. They use technology that makes operations safer and easier to do

You can have surgery to:

- take a sample of the tumour ([a biopsy](#)) for further tests or to diagnose the [type of brain tumour](#)
- remove all of the tumour (complete resection)
- remove as much of the tumour as is safe and possible (partial resection or debulking)
- reduce pressure in the skull.

To remove part or all of the tumour, you have an operation called a craniotomy. We have more information about [having a craniotomy and how it is done](#).

Occasionally, some people may also have [chemotherapy](#) given directly into the brain during surgery. If this would be helpful in your situation, your doctor will explain this to you.

Some tumours cannot be removed with surgery. A tumour may be too difficult to reach, or the risk of damaging some healthy parts of the brain could be too high. Your surgeon will talk to you about other treatment options.

Surgery to reduce pressure in the skull

Surgery may be needed to drain a build-up of fluid within the brain. This may happen if there is a blockage stopping the fluid from draining to another area of the body. This is called hydrocephalus.

If a brain tumour blocks the flow of fluid (CSF) around [the brain](#), pressure can build up and [cause symptoms](#). Treatments can include placing a shunt, or an endoscopic third ventriculostomy (ETV), or removing the tumour to unblock the blockage.

Shunts

A shunt is an operation to place a long, thin plastic tube into the brain. The shunt lets some of the fluid drain from the brain to another area of the body. Usually this is into the tummy (abdomen). It may make you feel better by reducing your symptoms.

You cannot see the shunt from outside the body. But you may be able to feel it under the scalp or in your abdomen. Your surgeon will explain what to expect. Sometimes shunts will be removed, or they can be left in for many years.

Endoscopic third ventriculostomy (ETV)

An endoscopic third ventriculostomy uses keyhole (endoscopic) surgery to make a small hole in the lining of the [ventricles in the brain](#). It lets fluid move past the blockage and reduces the pressure on the brain.

Other types of surgery

Another type of surgery may be used to place an Ommaya reservoir in the brain instead of a shunt.

Ommaya reservoir

If a tumour has a fluid-filled part (called a cyst), and the tumour cannot be removed, you may have an Ommaya reservoir placed into the cyst.

An Ommaya reservoir is a small hollow disc that sits under the skin of the scalp. It is connected to a tube that goes into the fluid-filled part of the tumour. If pressure builds up after the operation, your doctor can put a small needle through the skin into the disc and drain some fluid from the tumour. After it is in place, it can be drained more than once if needed.

Before your operation

You will have tests before surgery to make sure you are well enough to cope with it. These are usually done a few days before your operation at a pre-assessment clinic. They may include blood tests, heart and lung tests.

If you smoke, try to stop or cut down. This will help reduce your risk of problems, such as a chest infection. It will also help your wound to heal after the operation. Your GP can give you advice and support to help you stop smoking. We have more information about [stopping smoking](#).

If you are not already taking them, you may be given [steroids](#). They help to reduce swelling caused by the tumour. Always take steroids exactly as your doctor or nurse has prescribed. You usually take them for a while before and after surgery.

Talk to your doctor before your surgery if you take the following drugs:

- aspirin
- clopidogrel
- non-steroid anti-inflammatory drugs such as ibuprofen.

These drugs can thin the blood and increase the risk of bleeding after surgery. If you are on any blood-thinning medication, talk to your doctor before your surgery. They will give you advice about when you can take them.

You are usually admitted to hospital on the morning of your operation. Sometimes you will be admitted the day before.

You will meet a doctor from the surgical team and a specialist nurse who will talk to you about the operation. You will also meet the doctor who gives you the anaesthetic (anaesthetist).

Your feelings

Any operation to the brain is major surgery. It is natural to feel worried and frightened about it. You may have questions about:

- the risks of surgery
- whether the operation will change how your brain works, such as how you behave
- how you will feel after the operation
- how you will look, and whether you will have scars.

Your surgeon and specialist nurse will talk to you about the operation and the possible [benefits and risks](#). If there is a risk of damage to the brain during surgery, they will explain how this may affect you.

Your hair may sometimes be shaved during the operation. Some people find the thought of waking up with part of their head shaved very upsetting. If you are worried about this, your specialist nurse or surgeon can explain what to expect.

Your specialist nurse can give you and your family support. Make sure you have all the information you need. Talk about any concerns and ask any questions you have. Knowing what to expect can make it easier to cope and less frightening. You can also call the Macmillan Support Line for free on [0808 808 0000](tel:0808 808 0000)

After your operation

When you wake up, you will be in the recovery unit. After this you usually go back to your ward. Sometimes you may go to the intensive care ward or high-dependency unit for about 24 hours.

The doctors and nurses will monitor you carefully. They will:

- do checks, such as testing your reflexes and seeing how your eyes react to light
- ask questions and ask you to do things, to check you understand what is happening and can follow instructions
- take your temperature and blood pressure.

Your face and eyes may be swollen and bruised. This swelling and bruising should improve within a few days. Sometimes a swelling filled with fluid develops under the operation scar. This is called a meningocele. This may take week or months to go down, but it will get better over time.

Drips and drains

You may have some tubes in place when you wake up. These are used to give you fluid and drugs, or to drain body fluids. They are not usually painful. Tell your nurse if you are uncomfortable. Your nurse will remove each drip or drain when you no longer need it.

You may have some of the following:

- A drip going into a vein to give you fluids until you can eat and drink again.
- A drain from your wound to drain blood or fluid into a bottle. It is usually removed 1 to 2 days after the surgery.
- A fine tube that passes down the nose and into the stomach. This is called a nasogastric tube. It removes fluids from the stomach, to stop you being sick.
- A catheter. This is a tube that drains urine from your bladder. It is usually taken out when you can move around more.

Pain

You may have a headache when you wake up after the operation. The nurses will give you regular painkillers. Headaches usually get better after a few days. Always tell your nurse or doctor if you have pain, or if the pain gets worse.

Moving around

You will be encouraged to get out of bed as soon as you feel able. This is important to help prevent chest infections and blood clots. It also helps with your recovery. Your nurse can check that you are ready to get up. A physiotherapist or nurse will help you to start moving around if needed. You will usually be given special stockings to reduce the risk of developing a clot.

Your wound

The wound on your head may be covered with a dressing or bandage for the first few days. The nurses will check it regularly to make sure it is healing well. After about 7 to 10 days, they will remove your staples or stitches. This can be done at the hospital, at your GP practice or at home by a district nurse. If you have dissolving stitches, these will not need to be removed.

You can usually wash your hair around 3 days after your surgery with the medicated shampoo they give you. But try to keep the wound dry.

Once your stitches have been removed you can usually wash with your normal shampoo. Do not scrub the wound. Some people may find their wound feels tight. This is a normal feeling. It is important not to pick or pull the scab off, even weeks after the surgery.

Your doctor or nurse will explain how to look after the wound once you go home.

Recovery and going home

When you go home and how quickly you recover will depend on the type of operation you have. Your healthcare team during recovery may include:

- a physiotherapist
- an occupational therapist
- a speech and language therapist (SLT), if needed.

They can help you plan to go home and arrange any support you might need.

You will still be recovering when you leave hospital. Remember to take things slowly and follow the advice from your healthcare team. Contact the hospital straight away if you have any problems or new symptoms, which may include:

- a fever (high temperature) swelling, warmth or leaking from your wound.
- the wound looking red in people with white skin, or darker in people with black or brown skin
- feeling or being sick
- feeling very drowsy
- weakness in your arms or legs
- problems with speech
- a [seizure](#).

It is normal to feel [very tired](#) for several weeks or longer. For a few people, this may continue for 1 year or more. Getting enough rest and eating healthily will help you recover. Try to balance rest with some gentle exercise, such as regular short walks. This will help give you more energy.

Your surgeon and healthcare team will tell you what to expect and how you can help your own recovery. You can contact your clinical nurse specialist if you are worried about anything.

You will usually go back to the hospital a few weeks after your operation for a check-up and to discuss your results.

How a brain tumour may affect your right to drive

Following diagnosis and treatment for a brain tumour, most people will not be allowed to drive for a period of time. If you drive it is important to discuss with your doctor how your diagnosis and treatment for a brain tumour affects your right to drive.

If you have a driving licence, you must tell the licencing agency ([DVLA](#) or [DVA](#)) that you have been diagnosed with a brain tumour. We have more information about [how a brain tumour may affect your right to drive](#).

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary brain tumour information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

EANO-ESMO Clinical Practice Guidelines for prophylaxis, diagnosis, treatment and follow-up: Neurological and vascular complications of primary and secondary brain tumours. 2021. Available from www.eano.eu/publications/eano-guidelines/eano-esmo-clinical-practice-guidelines-for-prophylaxis-diagnosis-treatment-and-follow-up-neurological-and-vascular-complications-of-primary-and-secondary-brain-tumours [accessed August 2024].

NICE Guideline NG99. Brain tumours (primary) and brain metastases in over 16s. 2018 (updated 2021). Available from: www.nice.org.uk/guidance/ng99 [accessed August 2024].

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help